

**Year 8 Mathematics
Mini Test
Numbers and Indices**



Resources: one A4 page of notes

Time: 35 mins Name: So Lutions 32 / 32 marks

Full working out must be shown to get full marks. Attempt all the questions.

Part A – Directed Numbers

6 marks

Evaluate:

1. $-3 + 4 = 1$ ✓

2. $9 + -3 = 6$ ✓

(no half marks for this section)

3. $-17 - -3 = -14$ ✓

4. $-5 - 10 = -15$ ✓

5. $-5 \times 4 = -20$ ✓

6. $-8 \div -2 = 4$ ✓

Part B – Index Laws

16 marks

1. Complete the table (4 marks)

Index form	Expanded form
4^4	$4 \times 4 \times 4 \times 4$ ✓
7^3 ✓	$7 \times 7 \times 7$
5^5	$5 \times 5 \times 5 \times 5 \times 5$ ✓
$2^{\frac{1}{2}} \times 6^{\frac{1}{2}}$ ✓	$2 \times 2 \times 6 \times 6$

$\frac{1}{2}$ $\frac{1}{2}$

2. Simplify and leave the answer in index form. (1, 1, 2 marks)

a) $7^3 \times 7^4 = 7^7$ ✓

b) $10^5 \times 10^6 = 10^{11}$ ✓

c) $3 \times 3^2 \times 3^3 \times 4^3 = 3^6 \times 4^3$ ✓ ✓

3. Simplify and leave the answer in index form. (1, 1 marks)

a) $8^4 \div 8^2 = 8^2$ ✓

b) $\frac{6^5}{6^3} = 6^2$ ✓

4. Simplify and leave the answer in index form. (1, 1, marks)

a) $(5^2)^3 = 5^6$ ✓

b) $(8^{10})^2 = 8^{20}$ ✓

5. Evaluate the following. (1, 1, 2, 3 marks)

a) $154^0 = 1$ ✓

b) $5 \times 6^0 = 5 \times 1 = 5$ ✓

c) $2^2 \times 7 \times 4^0 = 2^2 \times 7 \times 1 = 4 \times 7 = 28$ ✓ ✓

d) $\frac{(4^2)^4}{4^5 \times 4^2} = \frac{4^8}{4^7} = 4$ ✓ ✓

Part C – Order of Operations**7 marks**

Solve (showing your working)

1. $2 + 7 \times -4 =$

$2 + -28 = -26$

~~2+7~~✓
✓

2. $(5 - 3 + 2)^2 + 4 \times -5 =$

$4^2 + 4 \times -5 =$

✓

$16 + 4 \times -5 =$

$16 + -20 = -4$

✓

3. $24 \div -2 \div -3 + 4 \times -2 - 5 =$

$-12 \div -3 + 4 \times -2 - 5 =$

$4 + 4 \times -2 - 5 =$

$4 + -8 - 5 =$

$-4 - 5 = -9$

End of Test

